

MINIMAL ANCIENT SOLUTIONS FOR THE NAVIER–STOKES EQUATIONS IN SELF-SIMILAR VARIABLES

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ABSTRACT. We prove a minimal-element reduction for bounded ancient solutions of the three-dimensional Navier–Stokes equations in backward self-similar variables. The result is formulated with explicit compactness and closedness hypotheses: if a nonempty time-translation invariant collection of bounded ancient solutions has a fixed local L^3 nonvanishing normalization and is closed under locally smooth limits, then it contains a nonzero solution of minimal L^∞ size. Consequently, a Liouville theorem for such minimal solutions implies the corresponding Liouville theorem for the whole collection. We also record a concrete uniformly L^3 -tight setting in which the required closedness is verified directly.

1. INTRODUCTION

We consider the incompressible Navier–Stokes equations in backward self-similar variables,

$$(1.1) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0,$$

on $\mathbb{R}^3 \times \mathbb{R}$. The variables are related to the usual ancient Navier–Stokes variables by

$$V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t), \quad \tau = -\log(-t), \quad t < 0.$$

The drift in (1.1) fixes the self-similar center and scale. The remaining symmetry relevant to a Type I blow-up limit is translation of the logarithmic time variable τ .

This paper isolates the compactness step which is needed before a rigidity argument can be applied. Suppose a family of bounded ancient solutions is stable under time translation and locally smooth limits, and suppose every solution in the family carries a definite amount of L^3 mass in a fixed ball at some time. Among all such solutions, one can minimize the usual $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ norm. The infimum is positive and is attained by a nonzero ancient solution. The proof has three components: time-centering of the nonvanishing normalization, compactness in the local C^∞ topology, and lower semicontinuity of the L^∞ norm.

The main theorem is deliberately a reduction theorem. It does not assert that the resulting minimal solution is zero. Rather, it states the precise PDE conclusion which any subsequent rigidity argument must rule out. In this form the statement uses only the standard compactness theory for bounded ancient Navier–Stokes solutions.

1.1. Relation with the preceding papers. We use the preceding papers in the series as inputs rather than reproving their conclusions. Paper 0 proves that a terminal singular point carries a positive pressure-normalized Caffarelli–Kohn–Nirenberg density and records the local Type I/Type II dichotomy [1, theorems “Singular points carry positive local CKN concentration” and “Local blowup-rate dichotomy”]. Paper 1 converts the Type I branch into a smooth bounded ancient solution of (1.1); the pressure gauges, compactness of the rescaled sequence, scale

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selection, non-triviality, renormalized equation, renormalized boundedness, and smoothness are established there [2, lemmas “Pressure gauges and local pressure bounds,” “Scale selection from local concentration,” “Renormalized equation,” “Renormalized boundedness and non-triviality,” “Smoothness,” and propositions “Compactness” and “Nonzero ancient limit”]. Paper 2 proves the bounded mild smoothing and compactness results used below, together with uniform L^3 -tightness closure and the stationary self-similar rigidity interface [3, lemmas “Smoothing of bounded mild solutions,” “Local smooth compactness,” “Closure of uniform L^3 -tightness,” theorem “Stationary limits under uniform L^3 -tightness,” and corollary “Stationary solutions under uniform L^3 -tightness”]. The last two results in Paper 2 invoke the stationary theorem of Nečas–Růžička–Šverák [9]; it is not reproved here.

Thus (3.1) is not a new concentration theorem. It is the abstract local L^3 normalization retained by the minimization argument after the pressure-normalized nonvanishing from Papers 0 and 1 has already been converted into a nonzero centered ancient solution. The only new argument in this paper is the order-completeness step: after time-centering a minimizing sequence, local smooth compactness and lower semicontinuity produce an element of minimal L^∞ -size.

1.2. What must be checked. The minimization argument uses four facts and no hidden compactness assumption. The class must be uniformly bounded in L^∞ , otherwise there is no compactness. It must contain a fixed local nonvanishing normalization, otherwise the minimizing sequence could converge to zero. It must be invariant under time translation, because the nonvanishing time of a given solution need not be $\tau = 0$. Finally, it must be closed under locally smooth limits which retain the normalization. These are precisely the four clauses in Theorem 3.1.

We use the following standard PDE conventions throughout. A locally smooth limit means convergence in $C^m(K)$ for every compact $K \subseteq \mathbb{R}^3 \times \mathbb{R}$ and every $m \geq 0$. Since this topology is generated by a countable exhaustion and the C^m -seminorms, all compactness arguments below may be written sequentially. All identities involving $\mathbb{P} \operatorname{div}$ are understood by duality against compactly supported smooth test functions, and then extended to weak- $*$ L^∞ identities by density in the relevant vector- or tensor-valued L^1 space. For a compact cylinder K , the notation $C^m(K)$ means the supremum norm of all mixed space-time derivatives $\partial_y^\alpha \partial_\tau^j$ with $|\alpha| + j \leq m$. The symbol B_R always denotes the ball in $\mathbb{R}^3$ centered at the origin.

2. BOUNDED ANCIENT SOLUTIONS

After applying the Leray projection $\mathbb{P}$, equation (1.1) takes the form

$$(2.1) \quad \partial_\tau V = \Delta V - \frac{1}{2} y \cdot \nabla V - \frac{1}{2} V - \mathbb{P} \operatorname{div}(V \otimes V).$$

Here $\mathbb{P} = I - \nabla \Delta^{-1} \operatorname{div}$ is the usual whole-space Helmholtz–Leray projection. Thus (2.1) is understood distributionally: for divergence-free test fields the pressure term in (1.1) drops out, and the pressure is recovered, up to a function of time, by the whole-space Riesz transform formula

$$-\Delta P = \partial_i \partial_j (V_i V_j).$$

No local pressure gauge is used below; the mild formulation fixes this whole-space Leray gauge. These are the standard whole-space conventions for the Navier–Stokes equations; see, for instance, Galdi [5] and Temam [11]. Let $S(a)$, $a > 0$, denote the semigroup generated by

$\Delta - \frac{1}{2}y \cdot \nabla - \frac{1}{2}$. It is given by

$$(2.2) \quad (S(a)f)(y) = e^{-a/2} \int_{\mathbb{R}^3} \frac{\exp\left(-\frac{|z-e^{-a/2}y|^2}{4(1-e^{-a})}\right)}{[4\pi(1-e^{-a})]^{3/2}} f(z) dz.$$

The kernel in (2.2) has mass $e^{-a/2}$. Consequently

$$\|S(a)f\|_{L^\infty} \leq e^{-a/2} \|f\|_{L^\infty}, \quad a > 0.$$

We shall also use the standard Oseen-kernel estimates, in self-similar variables,

$$(2.3) \quad \|S(a)\mathbb{P} \operatorname{div} F\|_{L^\infty(\mathbb{R}^3)} \leq C e^{-a/2} (1-e^{-a})^{-1/2} \|F\|_{L^\infty(\mathbb{R}^3)}$$

and

$$(2.4) \quad \|(S(a)\mathbb{P} \operatorname{div})^* \phi\|_{L^1(\mathbb{R}^3)} \leq C e^{-a/2} (1-e^{-a})^{-1/2} \|\phi\|_{L^1(\mathbb{R}^3)}$$

for $\phi \in C_c^\infty(\mathbb{R}^3)$. The singular factor is integrable at $a = 0$, and for fixed $a > 0$ the adjoint kernel is smooth and rapidly decaying when tested against a compactly supported smooth function. This is the usual heat–Stokes kernel bound transported through the Ornstein–Uhlenbeck change of variables; compare Fabes–Jones–Riviere [4], Giga–Miyakawa [6], and the semigroup notation of Pazy [10].

Definition 2.1 (Bounded mild ancient solution). A vector field $V : \mathbb{R} \rightarrow L^\infty(\mathbb{R}^3; \mathbb{R}^3)$, defined for every $\tau \in \mathbb{R}$ as a weak-* L^∞ time-slice, is a bounded mild ancient solution of (1.1) if

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^\infty(\mathbb{R}^3)} < \infty, \quad \operatorname{div} V(\tau) = 0$$

in distributions for every τ , if the scalar map

$$\tau \mapsto \langle V(\tau), \phi \rangle$$

is measurable for every $\phi \in L^1(\mathbb{R}^3; \mathbb{R}^3)$, and if for every $\sigma < \tau$

$$(2.5) \quad V(\tau) = S(\tau - \sigma)V(\sigma) - \int_\sigma^\tau S(\tau - s)\mathbb{P} \operatorname{div}(V \otimes V)(s) ds$$

as an identity in the weak-* topology of $L^\infty(\mathbb{R}^3)$. The norm of such a solution is

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} := \sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^\infty(\mathbb{R}^3)}.$$

The weak-* formulation means that (2.5) is interpreted after pairing with arbitrary $L^1(\mathbb{R}^3; \mathbb{R}^3)$ -test functions. In particular, the Duhamel integral is a weak-* integral: when paired with a test function, the resulting scalar integrand is understood to be Lebesgue measurable and the scalar integral is the corresponding absolutely convergent Lebesgue integral. For $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, this identity reads

$$\langle V(\tau), \phi \rangle = \langle V(\sigma), S(\tau - \sigma)^* \phi \rangle - \int_\sigma^\tau \langle V(s) \otimes V(s), (S(\tau - s)\mathbb{P} \operatorname{div})^* \phi \rangle ds,$$

where the last integral is absolutely convergent for $\phi \in C_c^\infty$ by (2.4). The Duhamel term is understood by this duality formula. It defines a bounded L^∞ -valued weak-* integral because (2.3) and $V \in L^\infty$ give

$$\int_\sigma^\tau \|S(\tau - s)\mathbb{P} \operatorname{div}(V \otimes V)(s)\|_{L^\infty} ds \leq C \|V\|_{L^\infty}^2 \int_\sigma^\tau e^{-(\tau-s)/2} (1 - e^{-(\tau-s)})^{-1/2} ds < \infty.$$

Equality against compactly supported smooth tests extends to equality against every $L^1(\mathbb{R}^3; \mathbb{R}^3)$ test function by density and boundedness of both sides in $L^\infty(\mathbb{R}^3; \mathbb{R}^3)$. This is the same

whole-space weak-* mild formulation used in Paper 2 [3]. After applying local smoothing, the weak-* time-slice representatives agree with the smooth representative on every compact subcylinder. All pointwise and local integral expressions below use this smooth representative. The term bounded mild solution on a time interval (τ_0, τ_1) has the identical meaning, with $\mathbb{R}$ replaced by (τ_0, τ_1) and with (2.5) required only for $\tau_0 < \sigma < \tau < \tau_1$.

Remark 2.2. The mild formulation fixes the whole-space Leray pressure, modulo functions of time. This excludes the parasitic pressure terms which may otherwise appear in local ancient solutions on the whole space.

Lemma 2.3 (Local smoothing). *Let V be a bounded mild solution of (1.1) on $\mathbb{R}^3 \times (\tau_0, \tau_1)$. For every compact cylinder*

$$K \Subset \mathbb{R}^3 \times (\tau_0, \tau_1)$$

and every integer $m \geq 0$,

$$\|V\|_{C^m(K)} \leq C(m, K, \tau_0, \tau_1, \|V\|_{L^\infty(\mathbb{R}^3 \times (\tau_0, \tau_1))}).$$

The same estimate holds uniformly for any family with a common L^∞ bound.

Proof. This is the bounded mild smoothing theorem proved in Paper 2, in the lemma *Smoothing of bounded mild solutions* [3, lemma “Smoothing of bounded mild solutions”]. That result is stated for the same equation (1.1), the same whole-space mild formulation, and the same L^∞ class. Its proof uses the self-similar change of variables to the usual Navier–Stokes system, the bounded-data local smoothing theory of Kato [7] and Giga–Miyakawa [6], and the standard interior parabolic bootstrap, in the classical form of Ladyzhenskaya–Solonnikov–Ural’tseva [8]. The cited statement records explicitly that the constants depend only on the derivative order, the compact cylinder, the time interval, and the displayed L^∞ bound, and therefore gives the asserted uniform family estimate under exactly the hypotheses stated above. $\square$

Remark 2.4 (Representatives and L^∞ norms). By Theorem 2.3, every bounded mild ancient solution has a smooth representative on $\mathbb{R}^3 \times \mathbb{R}$. Throughout the minimization argument we use this representative. For a continuous representative, the essential L^∞ norm agrees with the pointwise supremum: if $|V(y_0, \tau_0)| > \|V\|_{L^\infty} + \varepsilon$ at some point, continuity gives the same strict inequality on a small parabolic cylinder of positive measure, contradicting the definition of the essential supremum. Thus the pointwise estimates below are estimates for the stated $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ norm.

Proposition 2.5 (Local compactness). *Let V_n be bounded mild ancient solutions of (1.1) satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

Then a subsequence converges in

$$C_{\text{loc}}^m(\mathbb{R}^3 \times \mathbb{R})$$

for every $m \geq 0$ to a bounded mild ancient solution V of (1.1).

Proof. This is the local smooth compactness theorem proved in Paper 2, in the lemma *Local smooth compactness* [3, lemma “Local smooth compactness”]. The hypotheses there are identical: the sequence consists of bounded mild ancient solutions of (1.1) with a common L^∞ bound. The proof in Paper 2 uses the smoothing theorem *Smoothing of bounded mild solutions*, Arzela–Ascoli on a countable compact exhaustion, and weak-* passage to the mild Duhamel formula using the adjoint Stokes estimate. Hence it gives both the local C^m -subsequence convergence and the fact that the limit remains a bounded mild ancient solution. No pressure or far-field assumption beyond the whole-space Leray mild formulation is added. $\square$

3. CLOSED NORMALIZED COLLECTIONS

The only symmetry used below is translation in the time variable. For a solution V and $s \in \mathbb{R}$, write

$$V^s(y, \tau) = V(y, \tau + s).$$

If V is a bounded mild ancient solution, then V^s is again a bounded mild ancient solution with the same L^∞ norm. Indeed, for every $\phi \in L^1(\mathbb{R}^3; \mathbb{R}^3)$, the scalar map

$$\tau \mapsto \langle V^s(\tau), \phi \rangle = \langle V(\tau + s), \phi \rangle$$

is measurable because it is the composition of the measurable map $\rho \mapsto \langle V(\rho), \phi \rangle$ with the translation $\tau \mapsto \tau + s$. The distributional identity $\operatorname{div} V^s(\tau) = 0$ follows from $\operatorname{div} V(\tau + s) = 0$. If $\sigma < \tau$, then $\sigma + s < \tau + s$, and inserting these two shifted times in (2.5) gives

$$V^s(\tau) = S(\tau - \sigma)V^s(\sigma) - \int_{\sigma+s}^{\tau+s} S(\tau + s - q) \mathbb{P} \operatorname{div}(V \otimes V)(q) dq.$$

With the change of variables $q = r + s$, this becomes

$$V^s(\tau) = S(\tau - \sigma)V^s(\sigma) - \int_{\sigma}^{\tau} S(\tau - r) \mathbb{P} \operatorname{div}(V^s \otimes V^s)(r) dr.$$

The equality is an equality in the same weak-* L^∞ sense as (2.5). Thus the weak-* mild formulation is preserved. Finally,

$$\sup_{\tau \in \mathbb{R}} \|V^s(\tau)\|_{L^\infty(\mathbb{R}^3)} = \sup_{\tau \in \mathbb{R}} \|V(\tau + s)\|_{L^\infty(\mathbb{R}^3)} = \sup_{\rho \in \mathbb{R}} \|V(\rho)\|_{L^\infty(\mathbb{R}^3)},$$

so time translation preserves the L^∞ size.

Hypothesis 3.1 (Closed normalized collection). Let A be a collection of bounded mild ancient solutions of (1.1). Assume the following.

(1) There is $M < \infty$ such that

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M \quad \text{for every } V \in A.$$

(2) There are $R_0 > 0$ and $\eta_0 > 0$ such that

$$(3.1) \quad \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 \quad \text{for every } V \in A.$$

(3) If $V \in A$ and $s \in \mathbb{R}$, then $V^s \in A$.

(4) If $V_n \in A$, $V_n \rightarrow V$ locally smoothly on $\mathbb{R}^3 \times \mathbb{R}$, and V satisfies (3.1), then $V \in A$.

Remark 3.2. The constants R_0 and η_0 are part of the normalization. In a Type I blow-up construction they are obtained after the scale selection and nonzero centered-limit construction of Paper 1, which in turn uses the positive local CKN concentration supplied by Paper 0 [1, 2]. In this paper they are fixed data of the closed collection A , not extra assumptions on an individual minimizing sequence.

Lemma 3.3 (Time centering). *Let A satisfy Theorem 3.1. If $V \in A$, then for every $\delta > 0$ there is $s \in \mathbb{R}$ such that $V^s \in A$ and*

$$\int_{B_{R_0}} |V^s(y, 0)|^3 dy \geq \eta_0 - \delta.$$

Proof. The quantity

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy$$

need not be attained. The definition of the supremum nevertheless implies that, for every $\delta > 0$, there is $s \in \mathbb{R}$ such that

$$\int_{B_{R_0}} |V(y, s)|^3 dy \geq \eta_0 - \delta.$$

Then $V^s \in A$ by time-translation invariance, and the displayed inequality is the asserted one at $\tau = 0$. $\square$

Lemma 3.4 (Stability of the normalization). *Assume $V_n \rightarrow V$ locally uniformly on $\mathbb{R}^3 \times \mathbb{R}$. If*

$$\int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0 - \delta_n, \quad \delta_n \downarrow 0,$$

then

$$\int_{B_{R_0}} |V(y, 0)|^3 dy \geq \eta_0.$$

Proof. Restrict the convergence to the compact set $\overline{B_{R_0}} \times \{0\}$. Uniform convergence there implies

$$\|V_n(\cdot, 0) - V(\cdot, 0)\|_{L^3(B_{R_0})} \leq |B_{R_0}|^{1/3} \|V_n(\cdot, 0) - V(\cdot, 0)\|_{L^\infty(B_{R_0})} \rightarrow 0.$$

The functional $f \mapsto \int_{B_{R_0}} |f|^3$ is continuous on $L^3(B_{R_0})$. Indeed, if $f_n \rightarrow f$ in $L^3(B_{R_0})$, then

$$|\|f_n\|_{L^3(B_{R_0})} - \|f\|_{L^3(B_{R_0})}| \leq \|f_n - f\|_{L^3(B_{R_0})} \rightarrow 0,$$

and therefore $\|f_n\|_{L^3(B_{R_0})}^3 \rightarrow \|f\|_{L^3(B_{R_0})}^3$. Applying this to $f_n = V_n(\cdot, 0)$ gives

$$\int_{B_{R_0}} |V(y, 0)|^3 dy = \lim_{n \rightarrow \infty} \int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0.$$

$\square$

Lemma 3.5 (Lower semicontinuity of size). *Assume $V_n \rightarrow V$ locally uniformly on $\mathbb{R}^3 \times \mathbb{R}$. Then*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. If the liminf is infinite, the assertion is immediate. Choose a subsequence n_k such that

$$\|V_{n_k}\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \rightarrow L := \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

For every point (y, τ) , local uniform convergence gives pointwise convergence along this subsequence. Using the smooth representatives fixed in [Theorem 2.4](#), we get

$$|V(y, \tau)| = \lim_{k \rightarrow \infty} |V_{n_k}(y, \tau)| \leq \liminf_{k \rightarrow \infty} \|V_{n_k}\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} = L.$$

Taking the supremum over (y, τ) and using [Theorem 2.4](#) gives the asserted $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ inequality. $\square$

4. THE MINIMAL SOLUTION

Theorem 4.1 (Existence of a minimal ancient solution). *Let A be nonempty and satisfy Theorem 3.1. Set*

$$m = \inf \{ \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} : V \in A \}.$$

Then

$$0 < m \leq M,$$

and there exists $V_ \in A$ such that*

$$\|V_*\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} = m.$$

In particular $V_ \not\equiv 0$.*

Proof. Since $A \neq \emptyset$, the set of sizes in the definition of m is nonempty. Since every element of A satisfies the common bound M , this set is contained in $[0, M]$. Hence m is a finite real number and $m \leq M$. The local nonvanishing gives a positive lower bound. Indeed, for every $V \in A$,

$$\eta_0 \leq \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \leq |B_{R_0}| \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}^3.$$

The first inequality is (3.1). The second follows because B_{R_0} has finite measure and, for every fixed τ ,

$$\int_{B_{R_0}} |V(y, \tau)|^3 dy \leq |B_{R_0}| \|V(\cdot, \tau)\|_{L^\infty(\mathbb{R}^3)}^3 \leq |B_{R_0}| \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}^3.$$

Taking the supremum in τ preserves the inequality. Thus

$$m \geq \left(\frac{\eta_0}{|B_{R_0}|} \right)^{1/3} > 0.$$

By the definition of the infimum, for every n there exists $V_n \in A$ with

$$m \leq \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq m + \frac{1}{n}.$$

For each n , apply Theorem 3.3 with $\delta = 1/n$. Replacing V_n by the corresponding time translate keeps V_n inside A and does not change its L^∞ norm, since

$$\sup_{(y, \tau) \in \mathbb{R}^3 \times \mathbb{R}} |V_n(y, \tau + s)| = \sup_{(y, \tau) \in \mathbb{R}^3 \times \mathbb{R}} |V_n(y, \tau)|.$$

Thus we may assume that

$$\int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0 - \frac{1}{n}.$$

For these translated solutions the common L^∞ bound is still M , because time translation is an isometry of $\mathbb{R}^3 \times \mathbb{R}$ for the supremum norm. Theorem 2.5 applies because all V_n obey the common L^∞ bound M . Hence a subsequence, not relabeled, converges locally smoothly to a bounded mild ancient solution V_* . By Theorem 3.4,

$$\int_{B_{R_0}} |V_*(y, 0)|^3 dy \geq \eta_0.$$

Since the supremum in (3.1) is at least the value at $\tau = 0$, this implies that V_* satisfies (3.1). The closedness assumption in Theorem 3.1 therefore gives $V_* \in A$. Finally, by Theorem 3.5,

$$\|V_*\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} = m.$$

Since m is the infimum over all elements of A and $V_* \in A$, one also has $m \leq \|V_*\|_{L^\infty}$. The two inequalities give equality. The lower bound $m > 0$ shows that the minimizer cannot be the zero solution. $\square$

Corollary 4.2 (Reduction to minimal solutions). *Let A satisfy [Theorem 3.1](#). If an independent argument rules out every nonzero element of A which attains the minimum in [Theorem 4.1](#), then A is empty.*

Proof. This is the contrapositive of [Theorem 4.1](#). $\square$

5. A UNIFORMLY TIGHT SETTING

We record a concrete criterion for the closedness hypothesis. Fix positive constants M , L , R_0 , and η_0 , and let $\omega : [1, \infty) \rightarrow [0, \infty)$ be a function with

$$\lim_{R \rightarrow \infty} \omega(R) = 0.$$

No monotonicity of ω is assumed. Let A be the collection of all bounded mild ancient solutions of [\(1.1\)](#) satisfying

$$(5.1) \quad \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M, \quad \sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L,$$

$$(5.2) \quad \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R) \quad \text{for every } R \geq 1,$$

and the local nonvanishing condition [\(3.1\)](#).

Proposition 5.1 (Closedness under uniform tightness). *The collection A defined by [\(5.1\)](#), [\(5.2\)](#), and [\(3.1\)](#) satisfies [Theorem 3.1](#).*

Proof. The L^∞ bound and the local nonvanishing condition are part of the definition of A . Time translation preserves [\(1.1\)](#), [\(5.1\)](#), [\(5.2\)](#), and [\(3.1\)](#): after replacing $V(y, \tau)$ by $V(y, \tau + s)$, the suprema over $\tau \in \mathbb{R}$ are unchanged, and the preceding paragraph shows that the mild formulation is also preserved.

It remains to verify closedness. Let $V_n \in A$ and suppose

$$V_n \rightarrow V \quad \text{locally smoothly on } \mathbb{R}^3 \times \mathbb{R},$$

where V satisfies [\(3.1\)](#). By [Theorem 2.5](#), the bounded sequence (V_n) has a subsequence converging locally smoothly to some bounded mild ancient solution W . The original sequence already converges locally smoothly to V , so the same subsequence also converges locally smoothly to V . Hausdorff uniqueness of limits in the Fréchet topology C_{loc}^∞ therefore gives $W = V$. Thus V is a bounded mild ancient solution; in particular, the equation and the Leray mild formulation hold for V . [Theorem 3.5](#) gives the L^∞ bound in [\(5.1\)](#).

Paper 2, in the lemma *Closure of uniform L^3 -tightness*, supplies the qualitative closedness of uniform L^3 -tightness under locally smooth convergence [[3](#), lemma “Closure of uniform L^3 -tightness”]. The present class records the stronger fixed-modulus tail bound [\(5.2\)](#); the following monotone-convergence argument verifies that this specified modulus is also closed. The argument uses only the displayed inequalities for each fixed radius R , and not any monotonicity of ω . Indeed, since $\omega(R) \rightarrow 0$, [\(5.2\)](#) implies

$$\lim_{R \rightarrow \infty} \sup_n \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V_n(y, \tau)|^3 dy = 0,$$

which is precisely the qualitative uniform tightness hypothesis used in Paper 2. For the L^3 bound, fix $R < \infty$ and $\tau \in \mathbb{R}$. Local smooth convergence gives uniform convergence on $\overline{B_R} \times \{\tau\}$.

Hence $|V_n(\cdot, \tau)|^3 \rightarrow |V(\cdot, \tau)|^3$ uniformly on B_R , and the finite measure of B_R gives convergence of the integrals:

$$\int_{B_R} |V(y, \tau)|^3 dy = \lim_{n \rightarrow \infty} \int_{B_R} |V_n(y, \tau)|^3 dy \leq L^3.$$

The integrals on the left are monotone in R . By the monotone convergence theorem,

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)}^3 = \lim_{R \rightarrow \infty} \int_{B_R} |V(y, \tau)|^3 dy \leq L^3.$$

This holds for every fixed τ , and therefore

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L.$$

For the tail estimate, fix $1 \leq R < R' < \infty$. Then for every τ ,

$$\int_{R < |y| < R'} |V(y, \tau)|^3 dy = \lim_{n \rightarrow \infty} \int_{R < |y| < R'} |V_n(y, \tau)|^3 dy \leq \omega(R).$$

Here the equality follows from local uniform convergence on $\overline{B_{R'}}$. More explicitly, local uniform convergence gives $|V_n(\cdot, \tau)|^3 \rightarrow |V(\cdot, \tau)|^3$ pointwise on the bounded annulus $\{R < |y| < R'\}$. The common L^∞ bound gives the integrable dominating function $M^3 \mathbf{1}_{\{R < |y| < R'\}}$, and dominated convergence applies because the annulus has finite measure. The annuli $\{R < |y| < R'\}$ increase to $\{|y| > R\}$ as $R' \rightarrow \infty$. Applying monotone convergence again yields

$$\int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R).$$

Taking the supremum in τ proves (5.2) for V . The assumed local nonvanishing (3.1) supplies the final defining condition of A . Thus $V \in A$, which is the closedness clause in Theorem 3.1. $\square$

Corollary 5.2 (Minimal solution in the uniformly tight setting). *If the collection A in Theorem 5.1 is nonempty, then A contains a nonzero solution of minimal*

$$L^\infty(\mathbb{R}^3 \times \mathbb{R})$$

norm.

Proof. This follows from Theorem 5.1 and Theorem 4.1. $\square$

Corollary 5.3 (Interface with stationary rigidity). *Let A be the uniformly tight collection in Theorem 5.1. If every minimal element of A is stationary in τ , then A is empty.*

Proof. If A were nonempty, Theorem 5.2 would give a nonzero minimal element V_* . By assumption $V_*(y, \tau) = W(y)$. The defining bounds of the uniformly tight class give

$$\sup_{\tau \in \mathbb{R}} \|V_*(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and, for every $R \geq 1$,

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V_*(y, \tau)|^3 dy \leq \omega(R).$$

Since $\omega(R) \rightarrow 0$, these are exactly the global L^3 -bound and uniform tightness hypotheses required in Paper 2. Applying the stationary-tightness corollary there, *Stationary solutions under uniform L^3 -tightness*, to the bounded mild ancient solution V_* gives $V_* \equiv 0$ [3, corollary “Stationary solutions under uniform L^3 -tightness”]. This contradicts (3.1), which every element of A satisfies. $\square$

6. CONCLUSION

The result of this paper is the following precise reduction. Under the closedness, time-translation invariance, boundedness, and fixed local nonvanishing hypotheses stated above, any nonempty collection of bounded ancient solutions contains a nonzero solution of smallest $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ norm. Therefore a full Liouville theorem in such a collection is reduced to the exclusion of these minimal solutions. In the uniformly L^3 -tight setting, the hypotheses are closed under locally smooth limits, and stationary minimal solutions are already excluded by the stationary self-similar rigidity interface proved in Paper 2.

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